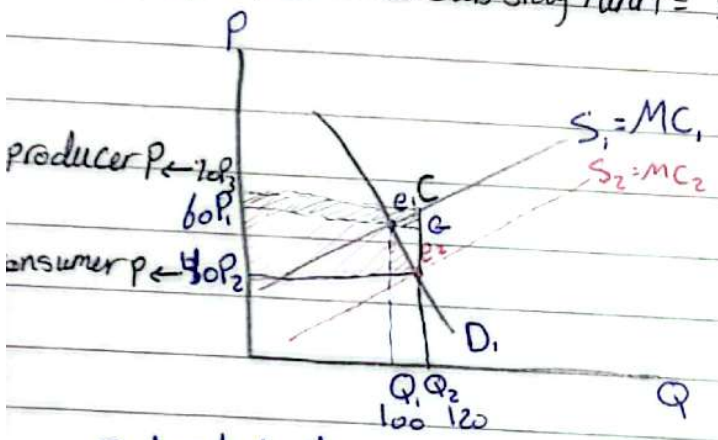
* transfer payment:

Ⓐ production subsidy:

ex a Commodity produced in a Competitive market, MC↑, ES > Ed.
$P_1 = 60$, $Q_1 = 100$. G subsidy/unit = 30. $Q_2 = 120$, $P_2 = 40$



- subsidy/unit = e_2C
- total cost of subsidy = $30 \times 120 = 3600 = P_3 P_2 e_2 C$
- benefit

Consumer: ↓ $P_1P_2 = 20 \times 120 = 2400$ = $P_1 P_2 E_2 G$

producer: ↑ $P_1P_3 = 10 \times 120 = 1200$ = $P_1 P_3 C G$

3600

* mainly to consumer (completely) مش

*Note: Negative relation between elasticity and benefit.
نص الاتمين X انخفاض السعر

- Δ in consumer surplus = $P_1 e_1 e_2 P_2 = \frac{1}{2}[100 + 120]20 = 2200$

مساحة شبه منحرف = ارتفاع × مجموع القاعدتين × ½

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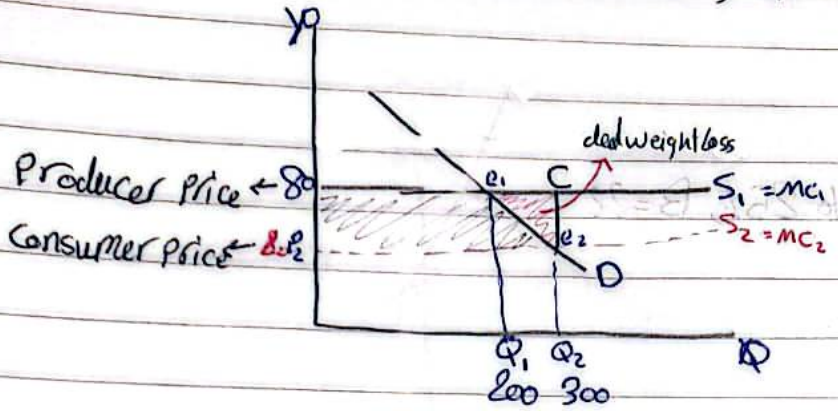
ex MC : constant, $E_s = \infty$, P_1 80, $Q_1 = 200$ G sub = 20 $P_2 = ?$ $Q_2 = 300$



• Benefit

consumer	producer
20 x 300	zero
6000	

• Δ in Consumer Surplus $= \frac{1}{2} \times [200 + 300] \times 20 = 5000$

B unemployment grant:

$$[B = G - t\,E]$$

B ↓ benefit, G ↓ grant, t ↓ reduction rate, E ↓ earned income ↓ W·H x wage/h

* break even incom → $E = \frac{G}{T}$

ex wage/h = 200, L.H = 10, W.H = 10, G = 600, T = 200

1- at L.H = 14 . E = ?

$E = 10 \times 200 = 200$

2- after sub, total I at L.h = 14 = ?

$E = 2000 + B = 600 - 20\% \times 2000 = 200$

Total income = 2200

3- to the level of earning at which he wouldn't recieve any benefit from grant

break even income $= \frac{600}{20\%} = 3000$

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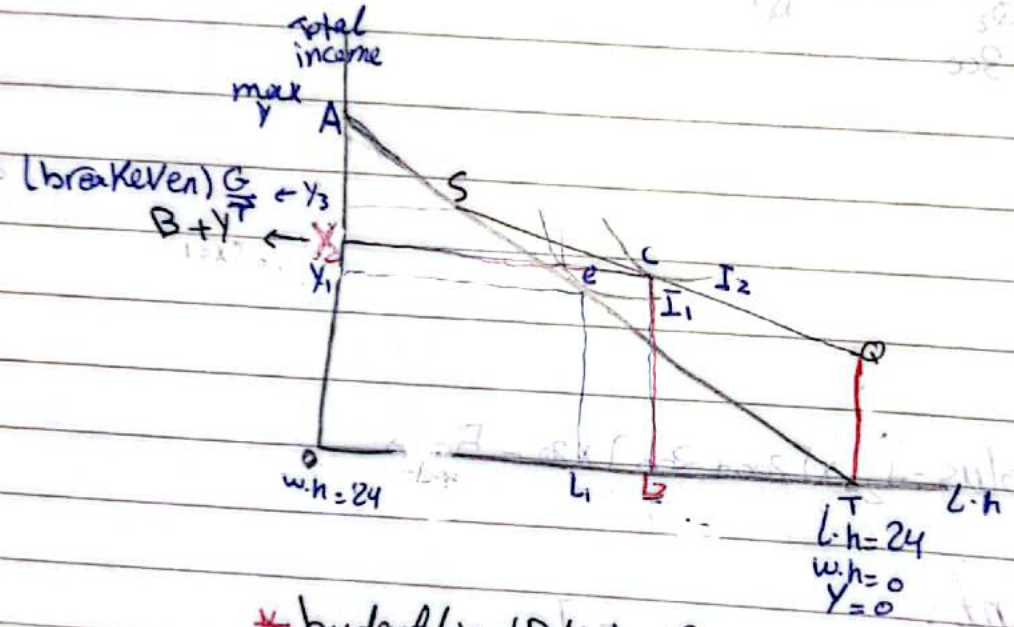
4- at breakeven income, $l.h = ?$

$w.h = \frac{\text{income}}{w.h} = \frac{3000}{200} = 15$

$l.h = 9.$

5- if the reduction rate increased to 25%, B = ??

$B = 600 - 25\% \times 2000 = 100.$

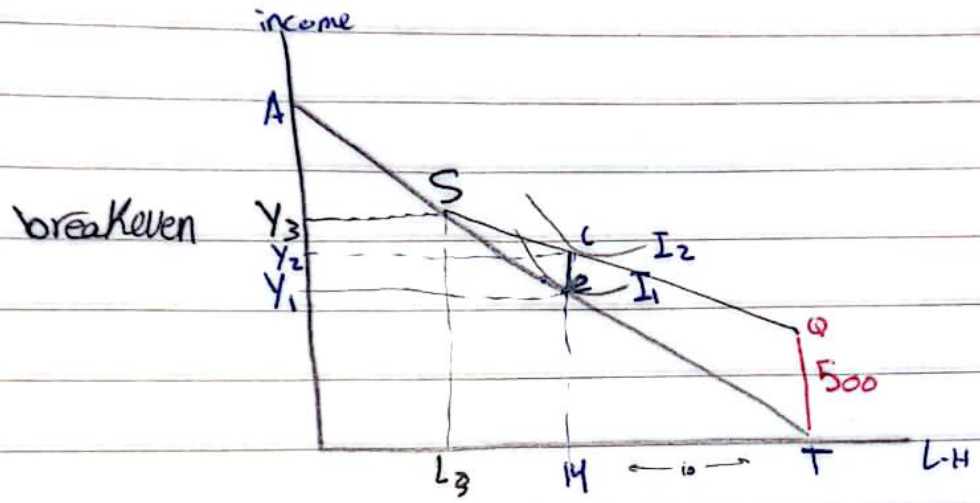


* budget line (BL) before grant = AT
* New BL = ASQ

* after $\begin{cases} l.H = 18 \\ w.H = 6 \end{cases}$

* $Y_2 = Y + R$

ex wage/h = 100 , L-h = 14 , w-h = 10 , G = 500 , T = 25%



- $Y_1 = w.h \times wage/h = 1000 \times 10 = 1000$

$Y_3 = \frac{G}{T} = \frac{500}{25\%} = 2000$

$Y_2 = 1000 + [500 - 25\% \times 1000] = 1250$

$L_3 = 4$